

# Week 6 — Law of Large Numbers, Duality, Conic Programming

AD7031 Optimization Under Uncertainty • Instructor: Prof. H. Park • Oct 3, 2026

## 1. Law of large numbers

$\tilde{\xi}_1, \tilde{\xi}_2, \tilde{\xi}_3, \dots$  : an infinite sequence of **independent and identically distributed (i.i.d.)** random variables,

each with expected value  $\mu$

**Law of large numbers** (strong law, SLLN)

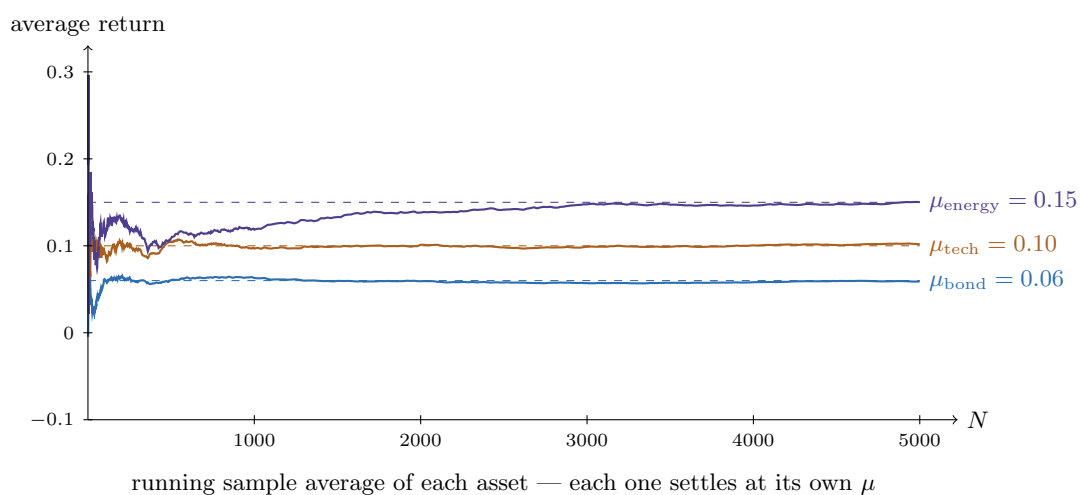
$$\mathbb{P}\left(\lim_{N \rightarrow \infty} \frac{1}{N} \sum_{i \in [N]} \tilde{\xi}_i = \mu\right) = 1$$

sample average  $\longrightarrow$  expected value

**Example** portfolio:  $\tilde{\xi}_i$  = return of an asset on day  $i$ , i.i.d. over days

| asset  | expected return $\mu$ | $\sigma$ |
|--------|-----------------------|----------|
| bond   | 0.06                  | 0.10     |
| tech   | 0.10                  | 0.20     |
| energy | 0.15                  | 0.30     |

$$\frac{1}{N} \sum_{i \in [N]} \tilde{\xi}_i = \text{average return over the first } N \text{ days}$$



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## 2. Duality theory

in this course, we focus on **convex** functions.

### Primal Problem

$$\begin{array}{ll} \inf & f_0(x) \\ \text{s.t.} & x \in \mathbb{R}^N \\ & f_i(x) \leq 0 \quad \forall i \in [I] \end{array} \quad (P)$$

### Dual Problem

$$\begin{array}{ll} \sup_{\theta} & \inf_{x \in \mathbb{R}^N} f_0(x) + \sum_{i \in [I]} \theta_i f_i(x) \\ \text{s.t.} & \theta \in \mathbb{R}^I \\ & \theta \geq 0 \end{array} \quad (D)$$

|                                                                                                                                |
|--------------------------------------------------------------------------------------------------------------------------------|
| $\begin{array}{ll} \inf & f_0(x) \\ \text{s.t.} & x \in \mathbb{R}^N \\ & f_i(x) \leq 0 \quad \forall i \in [I] \end{array}$   |
| $= \inf_{x \in \mathbb{R}^N} f_0(x) + \underbrace{\sup_{\theta \in \mathbb{R}_+^I} \sum_{i \in [I]} \theta_i f_i(x)}_{\leq 0}$ |

$$\begin{array}{ll} \text{if } f_i(x) \leq 0 \quad \forall i, & \text{then } = 0 \\ \text{if } \exists i : f_i(x) > 0, & \text{then } = +\infty \end{array}$$

$$\geq \sup_{\theta \in \mathbb{R}_+^I} \inf_{x \in \mathbb{R}^N} f_0(x) + \sum_{i \in [I]} \theta_i f_i(x)$$

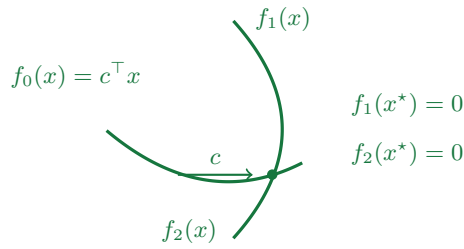
### Properties:

— **weak duality**      $\inf(P) \geq \sup(D)$      always holds.

— **strong duality:**      $\inf(P) = \sup(D)$  holds

when the problem (P) is convex and the Slater's condition holds

$$\exists x \in \mathbb{R}^N, \text{ s.t. } f_i(x) < 0 \quad \forall i \in [I]$$



In this course, we assume that strong duality holds for convex problem!!

### 3. Example: linear program

Primal LP:

$$\begin{aligned} \inf \quad & c^\top x \\ \text{s.t.} \quad & x \in \mathbb{R}^N \\ & Ax \leq b \end{aligned} \quad \begin{aligned} f_0(x) &= c^\top x \\ f_i(x) &= a_i^\top x - b_i \quad \forall i \in [I] \end{aligned}$$

$$\begin{aligned} f_i(x) &\leq 0 \\ \iff a_i^\top x - b_i &\leq 0 \\ \iff a_i^\top x &\leq b_i \end{aligned}$$

Dual LP:

$$\begin{aligned} & \sup_{\theta \in \mathbb{R}_+^I} \inf_{x \in \mathbb{R}^N} c^\top x + \sum_{i \in [I]} \theta_i (a_i^\top x - b_i) \\ &= \sup_{\theta \in \mathbb{R}_+^I} \inf_{x \in \mathbb{R}^N} c^\top x + \sum_{i \in [I]} \theta_i a_i^\top x - \sum_{i \in [I]} \theta_i b_i \\ &= \sup_{\theta \in \mathbb{R}_+^I} - \sum_{i \in [I]} \theta_i b_i + \underbrace{\inf_{x \in \mathbb{R}^N} \left( c + \sum_{i \in [I]} \theta_i a_i \right)^\top x}_{= \begin{cases} 0 & \text{if } c + \sum_{i \in [I]} \theta_i a_i = 0 \\ -\infty & \text{if otherwise.} \end{cases}} \\ &= \sup_{\theta \in \mathbb{R}_+^I} -b^\top \theta \\ & \text{s.t.} \quad A^\top \theta = -c \end{aligned}$$

LP strong duality : if there exists  $x \in \mathbb{R}^N : Ax \leq b$

$$\text{or } \theta \in \mathbb{R}^I : A^\top \theta = -c, \theta \geq 0$$

then strong duality holds

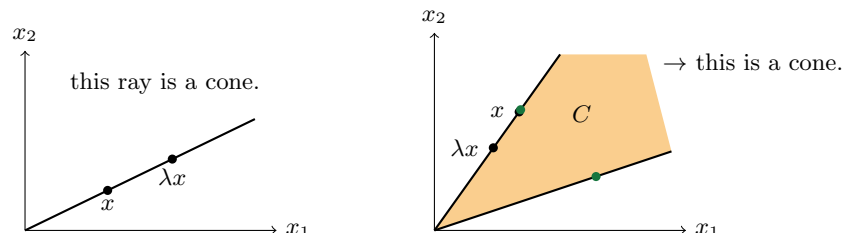
## 4. Conic programming

a conic program is a convex optimization problem.

— a **linear** objective function.

— a feasible set defines through the intersection of halfspaces  $Ax \leq b$  or a proper cone.

**Def (cone)** a set  $C$  is a cone if  $\forall x \in C$ , we have  $\lambda x \in C \quad \forall \lambda \geq 0$



**Def (proper cone)**  $\mathbb{R}^N$  is a cone but not pointed.

$C$  is a proper cone if it's **convex, closed, pointed**

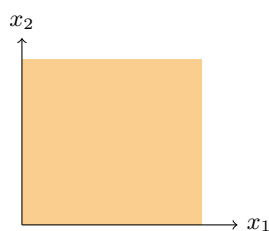
if  $x, -x \in C$  then  $x = 0$ .

**Generic conic program:**

$$\begin{aligned} \inf \quad & c^\top x \\ \text{s.t.} \quad & x \in \mathbb{R}^N \\ & Ax \preceq_c b \end{aligned} \quad \Longleftrightarrow \quad b - Ax \in C$$

**Example:**

1) **Linear program:**  $C = \mathbb{R}_+^K$



$$\begin{aligned} b - Ax &\in \mathbb{R}_+^K \\ \Longleftrightarrow b - Ax &\geq 0 \Longleftrightarrow Ax \leq b \end{aligned}$$

2) **Semidefinite program:**  $C = \mathbb{S}_+^K$

$\mathbb{S}^K$  : set of symmetric matrices in  $\mathbb{R}^{K \times K}$  ( $A^\top = A$ )

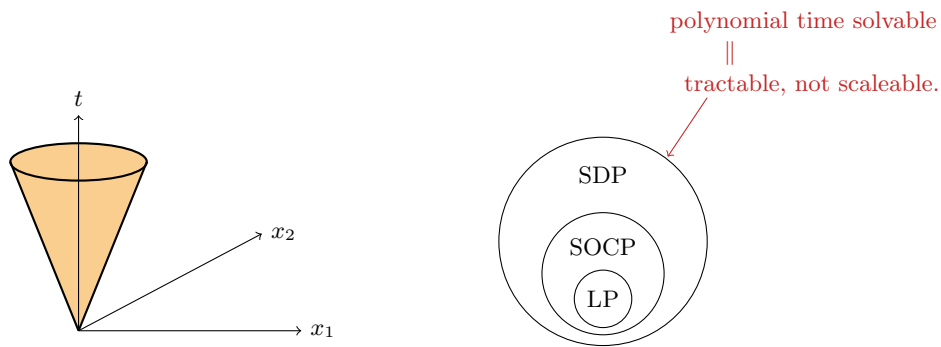
$\mathbb{S}_+^K$  : set of positive semidefinite (symmetric) matrices in  $\mathbb{R}^{K \times K}$

$A \in \mathbb{S}_+^K$  if  $z^\top A z \geq 0 \quad \forall z \in \mathbb{R}^K$

**Generic semidefinite program:**

$$\begin{aligned} \inf \quad & c^\top x \\ \text{s.t.} \quad & x \in \mathbb{R}^N \\ & A_1 x_1 + A_2 x_2 + \dots + A_N x_N \preceq_{\mathbb{S}_+^K} B \end{aligned} \quad \rightarrow \text{usually ignored}$$

3) **Second-order cone:**  $C = \{(x, t) \in \mathbb{R}^N \times \mathbb{R} : \|x\|_2 \leq t\}$  “ice-cream” cone



## 5. Conic programming duality

**Def** (*dual cone*)

$$C^* = \{y \in \mathbb{R}^K : \langle y, x \rangle \geq 0 \quad \forall x \in C\}$$

**Self dual cones** ( $C = C^*$ )

—  $C = \mathbb{R}_+^K$

—  $C = \mathbb{S}_+^K$

—  $C = \text{second-order cone}$

**Primal conic program:**

$$\begin{aligned} \inf \quad & c^\top x \\ \text{s.t.} \quad & x \in \mathbb{R}^N \\ & Ax \preceq_c b \end{aligned}$$

**Dual conic program:**

$$\begin{aligned} \sup \quad & -\theta^\top b \\ \text{s.t.} \quad & \theta \in C^* \\ & A^\top \theta = -c \end{aligned}$$